

Data Models

Summary Notes with Visual Structure

Definition

Data Model: A collection of tools to describe:

- **Data** – the entities and values stored.
- **Data Relationships** – associations among data items.
- **Data Semantics** – meaning of the data.
- **Data Constraints** – rules restricting possible data values.

Note

Data models provide an abstract representation of the real world for database design and implementation.

Types of Data Models

Overview

1. **Relational Model**
2. **Entity-Relationship (E-R) Model**
3. **Object-Based Data Models** (Object-Oriented, Object-Relational)
4. **Semi-Structured Data Model** (XML, JSON)
5. **Historical Models** (Network, Hierarchical)

1. Relational Model

Definition

A lower-level, record-based model that uses **tables (relations)** to represent both data and relationships. Each table consists of:

- **Columns:** represent attributes with unique names.
- **Rows:** represent records (tuples) of a particular type.

Note

- The relational model is the most widely used data model.
- Provides a structured, tabular representation of data.

Example

Example Table:

Student_ID	Name	Major
101	Alice	CS
102	Bob	EE
103	Carol	ME

2. Entity-Relationship (E-R) Model

Definition

A higher-level data model representing the real world as:

- **Entities:** objects distinguishable from others.
- **Relationships:** associations among entities.

Note

Primarily used for **database design**. Often converted into relational schemas for implementation.

3. Object-Based Data Models

- Object-Oriented Data Model:** Extends E-R model with encapsulation, methods (functions), and object identity. Common in Java, C++, C#.
- Object-Relational Data Model:** Combines features of object-oriented and relational models.

4. Semi-Structured Data Model

Definition

Allows data items of the same type to have **different sets of attributes**, unlike relational model where attributes are fixed.

Example

Common representations: **XML** and **JSON**.

5. Historical Models

Definition

Older models that precede the relational model; now rarely used:

- **Network Data Model**
- **Hierarchical Data Model**

Comparison Table

Model	Level	Structure	Use Case
Relational	Low	Tables/Rows	Widely used in DBMS
E-R	High	Entities & Relationships	Database design
Object-Oriented	High	Objects, Methods	OOP-based DBs
Object-Relational	Medium	Objects + Tables	Hybrid DBs
Semi-Structured	High	Tree/Graph	XML, JSON data
Network	Low	Graph (links)	Legacy DBs
Hierarchical	Low	Tree	Legacy DBs